

RL4Life: A Reinforcement Learning – based Method for Extending LEO Satellite Lifespan

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日期: 2024/11/14

Outline

- *Introduction & Motivation*
- *RL4Life Mechanism Overview*
 - *Contributions*
 - *Why Using RL*
- *Problem Formulation*

High Operating Costs

- **High Launch Costs:** **17 to 18 million** worth of Starlink satellites on each Falcon 9 launch [1].
- **Short Satellite Lifespan:** A Starlink satellite has a lifespan of approximately **5 years** [2].
- The rapid depletion of satellite node battery energy due to **overload** in the network will shorten the battery life cycle and severely reduce the satellite operational life [3].



Smart energy savings: maximize solar and battery power onboard



Cutting energy use eases system load and extends battery life



Switch off the radio channel when not in use

[1] <https://www.space.com/space-and-the-international-imperative-to-achieve-low-launch-cost/>

[3] F. Chen, Q. Wang and Y. Ran, "Dynamic Routing Algorithm for Maximizing Battery Life in LEO Satellite Networks," 2022 IEEE 8th International Conference on Computer and Communications (ICCC), Chengdu, China, 2022

RL4Life Mechanism Overview

- The LEO satellite installs the RL model, to determine the channel sleep according to the current traffic dynamic. (channel scheduling)

□ The LEO MUST turn off the channels as much as possible, and guarantee the user Quality of Service (QoS).

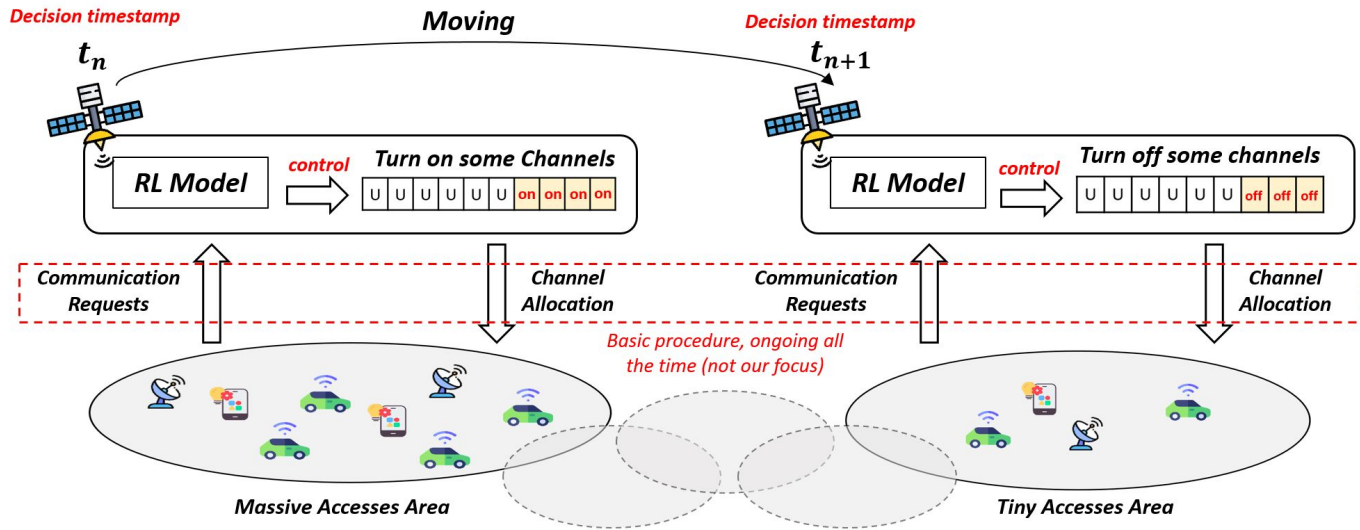


Fig. 2. RL4Life Mechanism Overview

RL4Life Mechanism Overview

Contributions

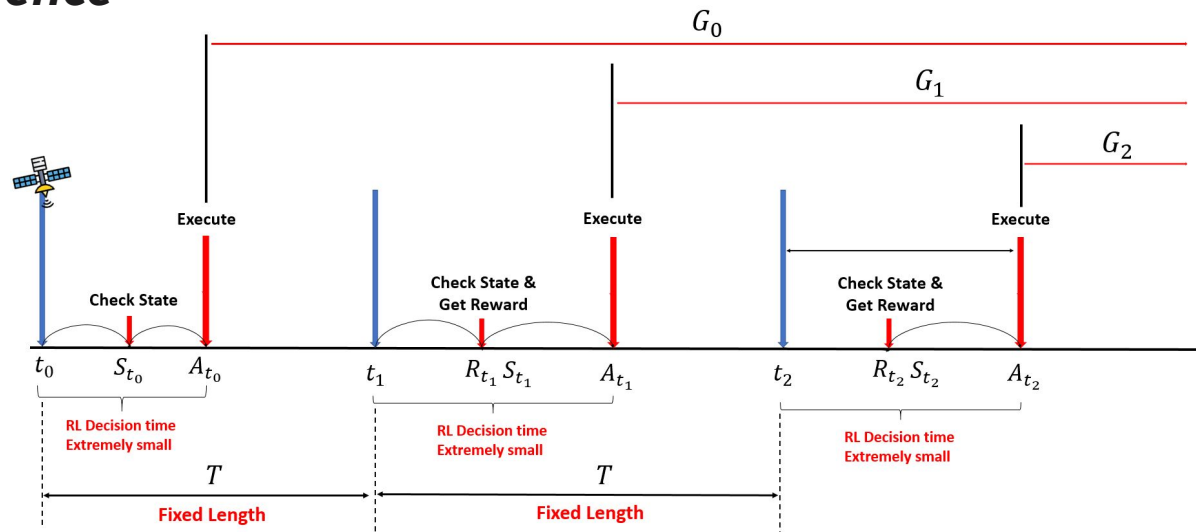
- ① We consider the mobility of the satellite and the traffic dynamic.
- ② We build up a simulation environment and finish the performance analysis to guarantee the correctness of the simulation.
- ③ Our RL4Life mechanism can significantly reduce energy consumption, and bring inspiration to reducing the *High Operating Costs* of the LEO satellite network.

Why Using RL

- Computation Cost
 - Satellites have limited computing resources, so it's **best to avoid external processing** on the satellite. Our plan is to install a pre-trained model directly on the satellite.
 - If optimization is used, large state space will be time-consuming.

Problem Formulation

Control Sequence



R.V. State S_{t_n} : $S_{t_n} = (\text{User_Matrix}_{t_n}, \text{OnWorking_Channel}_{t_n})$

Continue

Continue

The action will affect the state transition

R.V. Action A_{t_n} : $A_{t_n} = (a_{idle,1,t_n}, a_{idle,2,t_n}, \dots, a_{idle,m,t_n})$

$a_{idle,m,t_n} \in (0,1)$, mean m -th idle channel at t_n

R.V. Reward R_{t_n} $R_{t_n} = QoS(T) = \alpha_1 \times \text{Energy}_{t_n} + \alpha_2 \times \text{Channel_Utilization}_{t_n} + \alpha_3 \times \text{Switch_Cost}_{t_n}$

Unit: Watt

Unit: %

Penalty

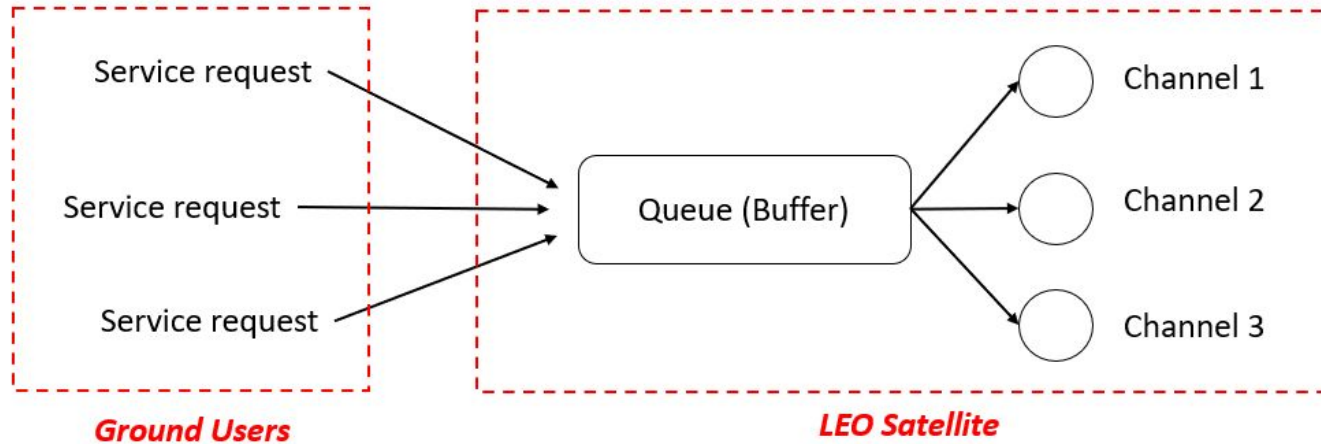
11/18 Meeting Outline

- *Recap*
- *Notations*
- *Simulation Environment (1)*

Simulation Environment (1)

System Behavior

- *User request service to LEO satellite -> It is Queueing Process*

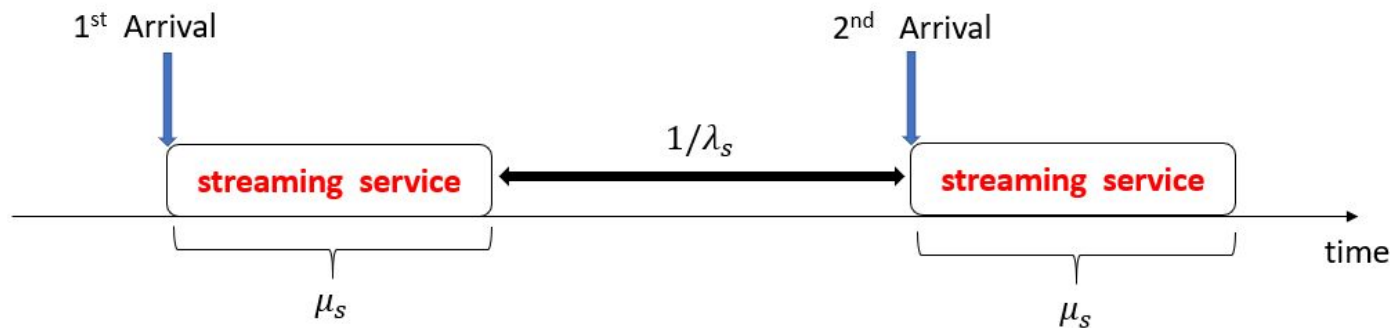


- *Our simulation environment will be a queueing system*
 - *Multiple Servers*
 - *Multiple Arrival Processes*

Simulation Environment (1)

Network Traffic Pattern

- We consider $N_{app} = 3$ applications services (i.e., voice call, streaming, browsing, etc).
- We assume the service arrival process is the Poisson Process with the arrival rate $\lambda_v, \lambda_s, \lambda_b$ (1/s), respectively.
- We assume the service dwelling time follows the exponential distribution with mean μ_v, μ_s, μ_b (seconds).



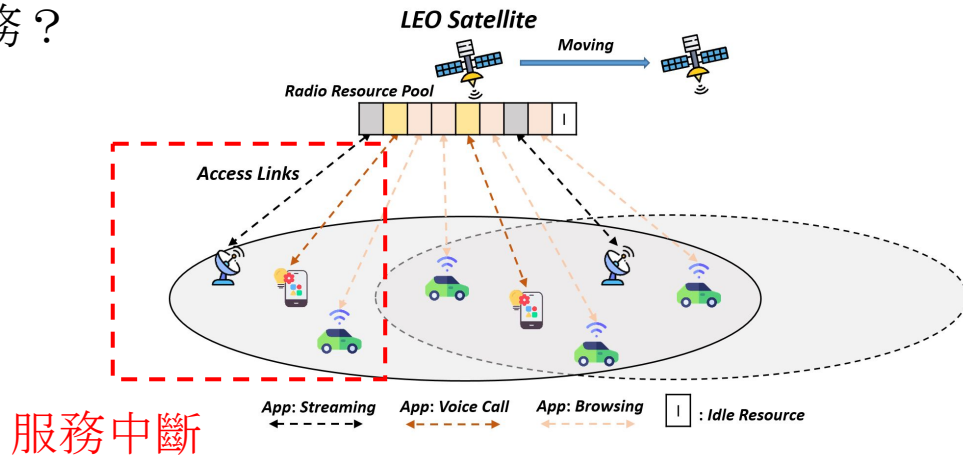
➤ The $\lambda_v, \lambda_s, \lambda_b$ are also the R.V.s

λ_v 數值的大小代表人數的多少 → 背後反映的是不同區域的忙碌程度。

LEO Satellite Mobility

- 衛星會移動;使用者正在使用服務,當衛星離開使用者的訊號範圍,服務會被中斷。
- 當服務中斷,如何處理被中斷的服務?
- 是否需要加入換手的行為?
- 中斷服務如何計入 Reward?

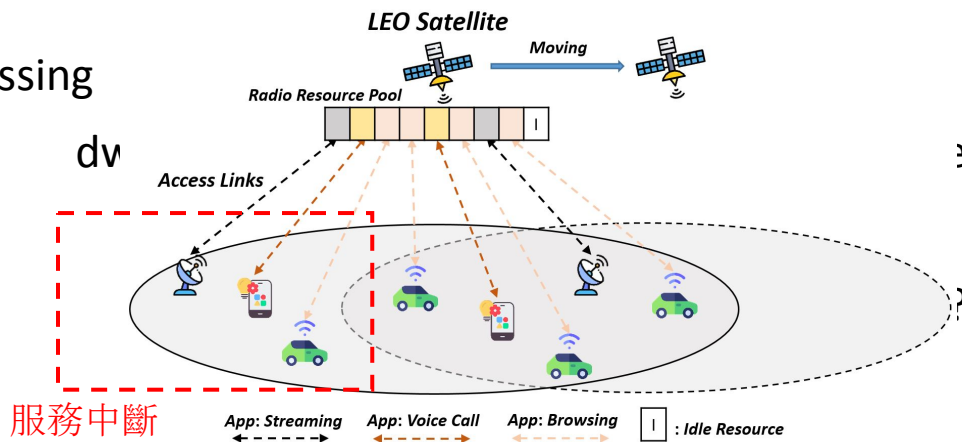
□ 會變複雜不少



Simulation Environment (1)

LEO Satellite

- Energy Model
 - Make channel sleep to save energy
 - Consumption : transmission & processing
 - longer the consumption is made
 - energy (recharge rate * exposure duration)

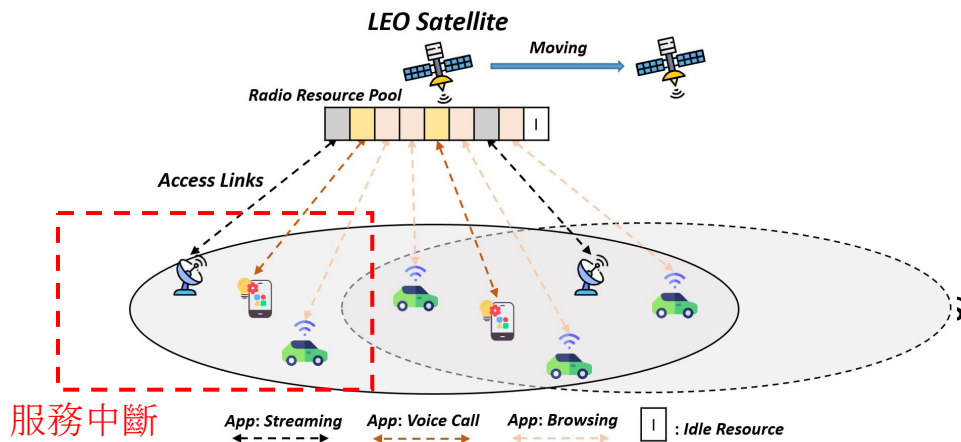


Simulation Environment (1)

LEO Satellite

- Time-Varying Channel Condition
 - transmission rate
 - remaining time slot for each user
- scattering UT
 - populations
 - different area

should



Simulation Environment (1)

Simulation Environment

- Hand-Craft
 - queueing model
 - build models from scratch? (e.g.
- [ns3-gym](#)
 - define network components and discrete events in NS3
 - difficult, not many tutorial out there
 - Can integrate with
- MATLAB (?)

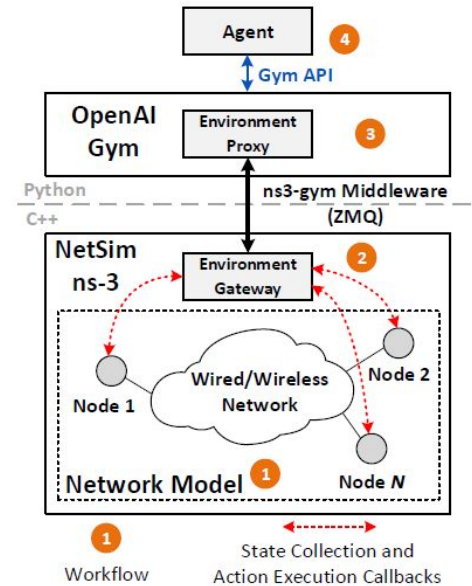


Figure 2: Architecture of ns3-gym framework.

Problem Definition

- link scheduling problem
 - a multi-beam sat. is to offload traffic from BS
- objective: maximize data/users served
- why use meta-RL
 - high complexity of optimal solution
 - resilient to dynamic env
- state space: channel coeff., power system, location & velocity of sat.
- action space: available link group; charge mode; downlink (if we want it complicate)
- we sample task from state and action space

Problem Definition

Meta-RL

- during meta training, diverse sampling is required, since it's infeasible to collect all possible environments
- to learn a meta critic trained in many diverse tasks
-

Simulation

- our system must adapt to variable env
 - traffic demand & wireless channel
- bursty demand: switch from low-demand voice call to high quality HD video service
- unseen channel variation
- recovering time metrics
 - time beginning from dramatic degradation to normal level

Comparison

- Heuristic method [\[1\]](#)
- MCTS